Tushar Gautam

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Work Experience

Buyhatke Internet Private Limited Bangalore

Technologies: JavaScript/TypeScript, Golang, Rust, Python, Express, NodeJS, Apache Kafka, MySQL, ClickHouse, Redis, WebSockets, Vector Embeddings/RAG, LLMs, Grafana, AWS

SDE-1 Backend *Aug. 2025 – Present*

- **Led development of a Hotel Comparison platform** integrating live rates from **7+ travel platforms** including MakeMyTrip, Booking.com, Agoda, Goibibo, Cleartrip, Expedia, and Kayak; built cross-provider hotel identity resolution and a **real-time WebSocket pipeline**, currently serving **16K+ compares/day** with **1.2K hourly peak** traffic.
- **Built and operate a live Flight Comparison platform** across **web and mobile**, processing **478K+ requests/day** and **3.79M+ result-served events/day**; engineered provider orchestration and quota-aware polling, with the Skyscanner integration recording **234K+ API calls/hour**, including **68K CREATE** and **54K POLL** calls.
- **Built a Food Comparison platform** across **Swiggy and Zomato**, using **FSSAI license numbers** for canonical restaurant identity matching and **ClickHouse** for high-throughput storage with asynchronous disk-spilled writes and live/fallback reads.
- **Designed a zero-LLM-in-request-path query categorization engine** matching **1.18M+ labeled examples** using multi-channel dense/sparse kNN retrieval and a **GBDT reranker**, achieving **ECE 0.0104** with calibrated confidence.
- Built the categorization hot path in **Rust** with mmap-loaded indexes, achieving **232ms cold start and p99 20ms** latency; maintained **¡ 0.1% cross-language disagreement** through Python–Rust–Node parity testing.
- Built an **LLM-powered semantic product-search system** using **Groq/OpenRouter, BGE-M3, and LanceDB**, with a self-improving cache and golden-benchmark evaluation harness, driving extraction accuracy to **99%**.

SDE Backend Intern *Jan. 2025 – Jul. 2025*

- **Led development of a cab comparison system** from scratch, aggregating **Uber, Ola, and Rapido** and building a smart-pickup feature that helped users save **1,000+ per month**.
- Engineered scalable **JavaScript web scrapers and backend APIs** for major e-commerce platforms and contributed to automated user-spend aggregation across **Amazon, Flipkart, Ajio, Myntra, and Meesho**.

Projects

Golang Enterprise Load Balancer Oct 2025

Multi-faceted HTTP load balancer with Redis sessions, circuit breakers, and full observability.

- Developed a high-performance **Go** load balancer with **weighted routing, circuit breakers, and Redis-backed sticky sessions** for fault tolerance.
- Enabled **observability and security** via **OpenTelemetry, Prometheus, mTLS**, and **rate limiting**; deployed cloud-natively with **Docker, Kubernetes, and Consul**.

VortexVault Nov. 2025

High-performance vector database with dual C++ and Rust backends for semantic search.

- Implemented a dual-backend architecture using **Crow (C++)** and **Actix-Web (Rust)** with identical REST APIs for performance benchmarking and a frontend dashboard for uploading files and querying.
- Containerized with **Docker Compose** for full-stack orchestration; supports persistent storage and backend switching at runtime.

Education

Indian Institute of Information Technology Kota GPA: 8.58/10

B.Tech, Electronics and Communication Engineering *Expected 2025*

Kautilya Senior Secondary School Score: 89%

CBSE – Class XII (PCM) *2020*

St. Paul's Convent School, Dasuya Score: 93.2%

ICSE – Class X *2018*